

An upstream placebo shows that reservoir siting explains the apparent drought buffering of seasonal reservoirs in Chile's megadrought

Francisco Zambrano Bigiarini¹

¹Facultad de Medicina Veterinaria y Agronomía, Universidad de Las Américas, Santiago, Chile

Key Points:

- Apparent drought buffering is as strong above dams as below, reflecting where seasonal dams sit, not what they do
- The powered null excludes large operational buffering; only carryover-capable reservoirs show an exploratory downstream-specific signal
- With storage stepping down as supply falls, expanding seasonal storage is supply-blind and adaptation must shift toward managing demand

Corresponding author: Francisco Zambrano Bigiarini, frzambra@gmail.com

Abstract

Much of what is attributed to reservoir drought buffering may be a measurable siting confound. We treat reservoir presence as a basin-level intervention in Chile’s megadrought and separate siting from operation with an upstream/downstream placebo. Across streamflow, the apparent buffering is as strong upstream as downstream, and it collapses once climate regions carry their own baseline transmission. Over the first decade of the megadrought the null is powered against large effects, excluding operational buffering above 46 to 59% of baseline transmission. But the fleet is storage-limited: in an exploratory split, the seven carryover-capable reservoirs (capacity at least half a year of downstream flow) show downstream-specific buffering that the seasonal majority does not; its direction is robust to the capacity classification and to single-basin exclusion, its significance to neither, so carryover reservoirs are candidates for genuine buffering rather than a confirmed exception. Dammed basins hold 4.5 times more cropland than matched controls, a siting legacy with no detected divergence within power; water-rights accrual is consistent with no effect but underpowered, a genuine null on new state grants that cannot see market reallocation in administratively closed basins. The storage band steps down at 2010 and the carryover floor keeps eroding. The upstream placebo is a portable method. For the storage-limited fleet, within these bounds, adaptation must shift toward managing demand.

Plain Language Summary

When drought strikes, governments often respond by building more reservoirs, on the assumption that storing water cushions communities when rainfall fails. But dams are built where water is already scarce, so basins with dams differ from basins without them even before any drought. We separated the effect of the dam from the effect of its location by comparing river gauges above and below each dam, and by matching dry dammed basins to similar undammed ones across Chile’s decade long megadrought. Once location is accounted for, the smaller, seasonal reservoirs that dominate the fleet show no measurable buffer against multi year drought, and the apparent buffering reflects where dams are built, not what they do. But the large carryover reservoirs, the few that hold at least half a year of flow, do show a downstream buffering signal, so the capacity to store water across years matters. With the water available to store falling, our results suggest that building more small reservoirs is not the answer; managing how water is used is.

1 Introduction

Storage is the reflexive response to drought, and the evidence that it is keeping up is weakening. Global reservoir capacity keeps growing through new construction, yet the fraction of that capacity actually holding water has declined (Li et al., 2023), and of 525 large reservoirs tracked by satellite altimetry between 2010 and 2022, 93% failed to fill at least once, with those in significant decline concentrated in the Southern Hemisphere between 14° and 44° S (Wang et al., 2024). Whether more storage is the right answer turns on how much drought relief a reservoir delivers, as opposed to appearing to deliver because of where it was built. Dams are not sited at random.

They are built where water is scarce and demand is high, often in direct response to the damage of a previous shortage (Di Baldassarre et al., 2018), under economic and political pressures that select particular basins (Kellner, 2021). Dammed basins are therefore drier and more water dependent than the undammed basins they are compared against before any drought begins, so the contrast mixes the effect of the dam with the reason the dam is there.

What reservoirs do at short timescales is reasonably well established. Operations shorten and soften hydrological drought at one to three month accumulations (Wu et al., 2017, 2018), and inflow-based counterfactuals find that regulation delays drought downstream (Wan et al., 2017; Wu et al., 2019, 2021). The multi-year timescale is where the evidence thins and the stakes rise. The same analyses that resolve short-term buffering report that the effect on 12 and 24 month drought is not apparent (Wu et al., 2018), dam drought resistance weakens or disappears as events intensify (Lee et al., 2026), and even a reservoir holding two years of mean annual flow mitigates decadal variability while leaving the rarest and deepest shortages largely untouched (Ho et al., 2017). Under structural aridification it is the multi-year deficit that decides whether storage protects anyone.

Settling the question requires a counterfactual, what the basin’s drought would have been without the dam, and each of the approaches in use leaves the siting problem partly open. Model-based and naturalized-flow reconstructions read regulation against simulated unregulated flow, but inherit the model’s memory, and conceptual storage models cannot accumulate deficits across years (Alvarez-Garreton et al., 2021), so the counterfactual is weakest at the timescale of interest. Before-and-after and regulated-versus-natural comparisons avoid the model but not the confound, since regulated and near-natural catchments are not distributed homogeneously across hydroclimatic zones, so climate imprints directly on the estimated regulation effect, as such studies note themselves (Götte & Brunner, 2024). Paired-gauge designs come closest, reading

a regulated reach against an unregulated reference (López-Moreno et al., 2009), and have been scaled nationally to flow-regime alteration (Chalise et al., 2021, 2023), but not to multi-year drought transmission conditional on the deficit each basin actually experienced. The field that framed the reservoir effect says as much: the tools to quantify the long-term dynamics of storage are still missing (Di Baldassarre et al., 2018).

We therefore build the counterfactual inside the basin. If a dam buffers drought, the transmission of meteorological deficit into streamflow deficit must flatten below the dam, where releases are managed, and not above it, where they are not. Reaches above and below the same dam share climate, geology, and the entire history that put a dam there, so differencing them removes siting without invoking a model, and if the attenuation is equally present upstream it is a property of the location rather than of the operation. This upstream/downstream placebo is our central identification strategy. We pair it with a matched cross-basin comparison against hydroclimatically comparable undammed basins, evaluating both against the drought each actually experienced rather than calendar time, which breaks the collinearity between megadrought onset and treatment. The estimand throughout is impact per unit of deficit, at annual to multi-year accumulation, the timescale of structural drought rather than of seasonal operation.

Chile is an acute testbed. Since 2010 its central regions have endured a *megadrought*, an uninterrupted run of dry years shading into structural aridification with a documented anthropogenic component (Boisier et al., 2018; R. Garreaud et al., 2025; R. D. Garreaud et al., 2017; Vicente-Serrano et al., 2019; Zambrano et al., 2025), and it sits in the band where reservoir levels are falling fastest worldwide (Wang et al., 2024). The 1981 Water Code makes water-use rights perpetual private property, so the policy reflex has been supply-side, new dams and inter-basin transfers, alongside market incentives that concentrated water in high-value export crops (Barría et al.,

2021; Reade Malagueño & D’Odorico, 2024; Rivera et al., 2016). In the snow-fed basins of central Chile demand already outruns supply in roughly three years of every ten (Webb et al., 2021). Chile thus exhibits every ingredient of the reservoir effect, making it a most-likely case for detecting one if it operates, and its short, steep Andean catchments, gauged above and below major dams, provide the physical setup the upstream placebo requires.

We adjudicate two accounts of the attenuated transmission that dammed basins have been reported to show, and bound what the data can decide between them. Under operational buffering the attenuation is a regulation effect, so it should appear below dams and not at the unregulated reaches above them, persist when dammed basins are compared within their own climate regime, and concentrate in the reservoirs with enough storage to carry water across years. Under siting it is a property of location, equally present upstream and dissolving once the aridity gradient along which reservoirs are sited is removed. The reservoir-effect hypothesis adds a demand-side prediction that runs against buffering altogether, that if storage induces the expansion it is built to serve, dammed basins should accumulate irrigated land and water claims faster through a prolonged drought. We test these predictions across the streamflow, cropland, and water-rights pathways with pre-specified falsifiers, and report the minimum detectable effect alongside every estimate, since a failure to reject is only as informative as the effect it can exclude.

2 Materials and Methods

2.1 Study area, units, and treatment

We study continental Chile (18–44° S), a strong north–south aridity gradient spanning hyper-arid, Mediterranean, and humid-temperate climates (Vásquez et al., 2025), over 2005–2024, when reservoir storage, meteorological forcing, and satellite outcomes jointly exist. The design is a matched comparison of dammed versus undammed sub-watersheds. Because most treated reservoirs (18 of 24) predate the storage record,

treatment is effectively time-invariant and a staggered-adoption event study is infeasible; identification rests on a matched cross-section conditioned on meteorological forcing rather than on calendar time. The estimand is a reservoir’s operational effect *conditional on the covariates that govern where reservoirs are sited*, isolating operation from *siting* (Di Baldassarre et al., 2018).

The analysis grain is the sub-watershed of Chile’s Dirección General de Aguas (DGA), the national water directorate, 467 units, chosen over the watershed grain because reservoirs sit in Chile’s largest main-stem basins and the finer grain supplies more controls inside the treated size range. A sub-watershed is treated if it contains 1 of the 26 monitored reservoirs, giving 24 treated units, of which 21 survive elevation common support. The reservoirs are predominantly irrigation (20 of 26), and the streamflow result is unchanged restricted to irrigation-only basins (Supplementary Table S1). Untreated sub-watersheds inside a reservoir’s watershed are removed as contaminated (45 units), leaving 398 clean controls, which defuses the most direct violation of the stable unit treatment value assumption, downstream spillover; no matched pair shares a watershed (Supplementary Table S2). Because “undammed” is defined against the 26 monitored reservoirs, the matched controls are additionally screened against the full 1,370-impoundment national inventory and the results re-fit with the entropy balance re-estimated on the dam-free control pool (Supplementary Table S3).

2.2 Data

2.2.1 Reservoir storage

Monthly storage for 26 monitored reservoirs (DGA, 2005–2026; we analyse complete calendar years 2005–2024) with per-reservoir capacity in hm^3 ; because storage and capacity share units, percent-of-capacity is the cross-reservoir-comparable state variable. The band trends are re-fit three further ways, since capacities span three orders of magnitude and several reservoirs enter the panel after 2005 with filling phases that

static fixed effects do not absorb, weighted by capacity and as fleet-total volumes (Supplementary Table S4), on the coverage-stable subset alone (Supplementary Table S5), and, because a descriptive trend offers no treatment assignment to permute, with every coefficient confirmed by a restricted wild cluster bootstrap using reservoir-level Rademacher weights (Supplementary Table S6). For the capacity heterogeneity test the buffering sample (17 treated basins with a usable downstream gauge) is split at a carryover ratio of 0.5, capacity divided by mean annual downstream flow in years. Because that denominator is regulated and consumed, the ratio is recomputed with unregulated upstream-gauge flow (Supplementary Table S7); and because the class mixes irrigation reservoirs with two dams operated also for hydropower (Colbún, Lago Laja), and seven units are few enough that one basin can move a permutation p across conventional thresholds, the contrast is re-estimated excluding that pair, under leave-one-out, and on strictly paired basins (Supplementary Tables S8, S9). That split is exploratory, carrying no multiplicity adjustment.

2.2.2 *Meteorological forcing*

The Standardized Precipitation Evapotranspiration Index (SPEI) at 12-month accumulation, from Climate Hazards Group Infrared Precipitation with Stations (CHIRPS) precipitation and Climate Hazards Center Infrared Temperature with Stations (CHIRTS) temperature standardized on 1991–2020 (Sergio M. Vicente-Serrano et al., 2010), is the exogenous dose; CHIRPS-derived indices are validated for Chilean drought monitoring (Meza, 2013; Oertel et al., 2020; Zambrano et al., 2017). The 12-month accumulation matches the annual, snowmelt-fed cycle of central Chile and the inter-seasonal carryover irrigation reservoirs bridge; estimates are insensitive to 6- and 24-month alternatives, and the window does not resolve 1 to 3 month buffering, a boundary of our conclusions (Wu et al., 2017, 2018). Conditioning on forcing breaks the collinearity between megadrought onset and treatment clock.

2.2.3 *Hydrological drought*

Daily streamflow from the DGA network (2000–2020), compiled by the Center for Climate and Resilience Research (CR2, <https://cr2.cl>), aggregated to the Standardized Streamflow Index at 12-month accumulation (SSI-12) (Lema et al., 2025; Sergio M. Vicente-Serrano et al., 2012), the regulated-flow analogue of SPEI-12 and the placebo outcome; gauges are classified upstream or downstream of each dam by Shuttle Radar Topography Mission (SRTM) elevation. Missing values are never imputed, monthly means require 20 valid days, the 12-month accumulation propagates any missing month, gauges need 96 monthly SSI values, and stations 10 years of record extending past 2010. Gaps therefore shrink the sample rather than distort the index, and their drought-conditional structure is quantified overall and by gauge class (Supplementary Table S10). SSI is standardized on the common 2000–2020 window so all gauges share one reference; because that window is megadrought-dominated and shorter than the 30-year guideline, estimates are re-derived on the 1990–2020 baseline, which the 1900-onset record permits (Supplementary Table S11). Because SSI normalizes each gauge, the slope could divide out dam-induced variance compression, so models are re-estimated on log 12-month mean flow and the compression premise tested directly (Supplementary Table S12).

2.2.4 *Land use, water rights, and auxiliary data*

Annual cropland area comes from MapBiomass Chile Collection 2 (30 m, 1999–2024) and evapotranspiration (ET) from the Moderate Resolution Imaging Spectroradiometer (MODIS) MOD16A2GF version 061 product (8-day, ~500 m, 2001–2024), whole-basin and for an orchard stratum from the national cadastre of Chile’s Natural Resources Information Centre and Office of Agricultural Studies and Policies; both track drought-driven agricultural losses in Chile (Zambrano et al., 2018). Consumptive rights from the DGA registry, which records each right’s grant date, flow, and capture point, are assigned to sub-watersheds as an annual cumulative stock; because the registry carries severe volume errors, the primary measure is rights density (count

per 100 km²), with a winsorized volume series insensitive to threshold (Supplementary Table S13) and rank-based tests on raw volumes (Supplementary Table S14; both in Supplementary Fig. S1). ERA5-Land reanalysis snow water equivalent (0.1°, 2000–2026) tests the snowmelt confound on the placebo estimand through a forcing-by-snowpack adjustment, a low-snow subsample, a snow-year interaction, and an early/late split (Supplementary Table S15); its bias is multiplicative, so standardization removes it, unlike the land-cover-dependent MOD16 ET bias that disqualifies the ET outcomes (Supplementary Table S16). DGA well hydrographs (2000–2021, 213 wells) give per-basin depletion rates bounding the groundwater-substitution confound (Supplementary Table S17).

2.2.5 Matching covariates

Baseline aridity (precipitation over potential evapotranspiration, P/PET, over the 1991–2020 World Meteorological Organization normal), watershed area, mean elevation (SRTM), and Köppen–Geiger class, by zonal statistics; recomputing aridity on the strictly pre-drought 1991–2009 window is essentially identical ($r = 0.998$), so the matching does not lean on drought-period variation (Supplementary Table S18).

2.3 Matched control construction

We estimate the average treatment effect on the treated (ATT) by *entropy balancing*, exact-matched within Köppen main group and balancing log(area) and mean elevation; latitude is left untargeted because it fights climate class across the Andes–Patagonia span, and balancing elevation raised the effective control sample from 9 to 91 while balancing latitude as a by-product. The procedure retains 21 of 24 treated units with essentially perfect balance (standardized mean differences falling from 1.18/0.37 to 0.000) against 244 in-window controls. Baseline aridity is a genuine confound (median P/PET 0.26 treated versus 0.80 control) but is *not* a balance target, because hard-balancing collapses the effective control sample from 91 to 19; the residual imbalance (standardized mean difference 0.17) is absorbed by

the doubly-robust outcome model, an interpolation within support since all treated basins lie inside the control aridity range (Supplementary Fig. S2), and weighting-only and overlap-subset fits leave the central nulls unchanged (Supplementary Tables S19, S20). Baseline cropland is deliberately not a balance target either, since cropland concentration is plausibly a consequence of siting and sits adjacent to the demand outcomes, so balancing it would condition on a mediator; its functional form is instead tested by metric invariance on log cropland share (Supplementary Table S21). Because transmission can be non-linear in aridity, the model is also re-fit with quadratic aridity-by-forcing interactions (Supplementary Table S22). The strict hard-balanced fit (log-aridity standardized mean difference 0.000, effective sample 19) is a primary sensitivity, reported with weight-concentration diagnostics and leave-one-out re-fits (Supplementary Tables S23, S24); at that sample it corroborates rather than powers the result. The decisive evidence, the within-basin placebo, the within-region comparison, and randomization inference, uses no outcome-model aridity adjustment, and coarsened exact matching and nearest-neighbour matching on the identical sample give the same effects.

2.4 Estimands, estimators, and inference

2.4.1 Matched ATT and transmission slope

Each outcome is estimated weighting only, regression only, and doubly-robust (entropy weights plus an outcome model in $\log(\text{area})$, elevation, and $\log(\text{aridity})$), so functional-form sensitivity is visible; covariates are centred at the treated mean and standard errors use the HC3 heteroskedasticity-consistent estimator. To remove megadrought *exposure*, concentrated in exactly the arid basins where reservoirs sit, the outcome for that machinery is the per-basin *transmission slope*, the regression of an annual outcome on annual SPEI-12; the vulnerability hypothesis predicts a *steeper* slope in dammed basins. Table 1 carries two level-change ATTs, cropland-area expansion (ha km^{-2} added) and water-rights accrual. A third outcome, reservoir ET buffering, applies the slope machinery to orchard-cell MOD16 ET and carries no evidential

weight, since MOD16 underestimates ET over irrigated land and biases that gap toward zero.

2.4.2 *Forcing-interacted difference-in-differences*

On the observed area and ET panels we fit

$$y_{it} = \beta (\text{dammed}_i \times \text{SPEI}_{it}) + \gamma \text{SPEI}_{it} + \alpha_i + \tau_t + \varepsilon_{it},$$

with sub-watershed (α_i) and year (τ_t) fixed effects and entropy weights. Because treatment is time-invariant the dose is meteorological SPEI rather than calendar year: year effects absorb the common megadrought shock and β is the differential deficit-to-impact slope, which the siting confound biases far less than it biases levels; the vulnerability hypothesis predicts $\beta > 0$. For slow-moving structural outcomes such as perennial cropland this tests only the annual drought-response differential, and cumulative expansion is instead tested by a dynamic event study (dammed $\times$ year, reference 2009) with bounded pre- and post-2010 trend tests. Because year effects absorb the common temporal component, β is identified from within-year spatial variation in SPEI; as a sensitivity retaining the temporal signal, the streamflow models are re-fit with year effects replaced by a cubic national time trend (Supplementary Table S25).

2.4.3 *Bounding the nulls*

Because the headline results are nulls, we bound the effect rather than rely on failure to reject: a two one-sided test declaring equivalence when the 90% interval lies within $\pm 25\%$ of baseline transmission, plus the threshold-free minimum detectable effect (MDE; 80% power, $\alpha = 0.05$). Since no hydrological standard defines a negligible reservoir effect, the margin is anchored in operational drought classification, whose severity categories are 0.5 standardized units wide (McKee et al., 1993; Sergio M. Vicente-Serrano et al., 2010), so a 25% slope change perturbs the propagated index

by less than one category, though it still translates to a materially non-trivial volume (Supplementary Table S26). MDEs and equivalence intervals use the permutation-null standard deviation, since cluster-robust standard errors over-reject at ~ 21 clusters. A positive control confirms the null is informative: injected buffering is recovered one-to-one, with detection once it exceeds the MDE.

2.4.4 Inference

With ~ 21 treated clusters, cluster-robust standard errors over-reject, so the primary inference is *randomization inference*: treatment is permuted among units within Köppen $\times$ aridity-tercile strata (999 permutations). Residual spatial dependence is assessed with Moran's I (Supplementary Table S27); where present, notably the water-rights outcomes, those outcomes are re-run with a permutation that swaps treatment at the watershed-block level, and this spatially valid inference governs where the two disagree. Residual confounding is probed with an aridity placebo relabeling the most-arid control tercile as pseudo-treated.

2.4.5 Siting-confound decomposition

A ladder on the same panel locates where any apparent buffering arises: (1) a naive pooled slope gap, what a conventional dammed-versus-undammed study reports; (2) unit and time fixed effects, unweighted; (3) the design estimator; (4) the design estimator with baseline transmission free to vary by Köppen region, reported with its own permutation null (Supplementary Table S28). The rungs are distinct controls, since the entropy balancing neither balances the residual within-region aridity gradient nor imposes a region-specific dose-response, so an apparent effect can survive rungs (1) to (3) and die at rung (4). The collapse from (3) to (4) is the between-region aridity confound; the residual at (4) is the regulation estimate.

2.5 Within-basin upstream/downstream placebo

The primary selection-versus-effect discriminator contrasts the SPEI-12 $\rightarrow$ SSI-12 transmission slope at regulated downstream reaches against unregulated upstream

reaches of the *same* basins, which share climate, geology, and selection history: a genuine regulation effect implies a downstream-specific dose-response the upstream placebo does not reproduce. Of the 21 treated matched basins, 17 have a usable downstream gauge and enter the buffering sample; 15 also have an upstream gauge and form the paired contrast. The contrast $D = \beta_{\text{down}} - \beta_{\text{up}}$ is tested against its permutation distribution (Supplementary Table S29). Because control basins have no dam and therefore no natural gauge partition, the scheme holds panel composition fixed: each panel contains the treated basins' gauges of that class plus every control basin's full gauge set; in each draw treatment is reassigned at basin level within Köppen $\times$ aridity-tercile strata and the *same* draw applied to both panels before D is recomputed. Under the sharp null gauge class carries no dam-specific information, so D is centred on zero, and one draw across both panels preserves the between-panel dependence. Because a pseudo-treated control contributes an identical gauge set to both panels, the up/down heterogeneity real treated basins carry is absent from its null contribution, which could compress the null variance and make the test anti-conservative; that bias can only overstate significance, never manufacture a null, and both contrasts are re-tested under a variant null partitioning each pseudo-treated basin's gauges at its median elevation (Supplementary Table S30). Subgroup and covariate-adjusted contrasts inherit the same construction, and the contrast is also split by Köppen region.

The placebo's comparability assumptions are measured rather than assumed. Upstream reaches sit higher and drain smaller areas, so we quantify the elevation and area gradients among undammed control gauges and re-fit the placebo net of a SPEI-by-elevation interaction and, using each gauge's mean annual discharge as the cumulative-scale measure the local polygon area misses, net of a SPEI-by-log-discharge interaction (Supplementary Tables S31, S32, S33, S34); the linearity of the elevation sensitivity is checked across the ~ 2000 m snowline (Supplementary Table S21) and the bedrock-to-alluvial transition with 2.5 km local relief (Supplementary

Table S35). Because gauge class is assigned from SRTM elevation, misclassification is bounded three ways: excluding downstream gauges whose DGA sub-cuenca is named for a different river and those within 100 m of the dam (Supplementary Table S36); validating against routed HydroRIVERS v1.0 topology, snapping every gauge and dam to its nearest reach, tracing the NEXT_DOWN path, and re-estimating on network-connected gauges only (Supplementary Table S37); and screening upstream gauges for cascade contamination against the national inventory at three tiers, below a *material* dam (DGA-monitored, registry-large, or wall height ≥ 15 m) inside their own sub-watershed, below the documented historical cascades, and below any material dam anywhere in their cuenca, the last over-flagging parallel tributaries to bound the worst case (Supplementary Table S38).

Four further specifications bound the remaining threats: a quadratic SPEI interaction and drought-month refits, testing whether the linear form masks an effect confined to extreme deficits (Supplementary Tables S39, S40); an early/late megadrought split, tested as a $\text{treat} \times \text{SPEI} \times \text{phase}$ interaction under permutation (Supplementary Tables S41, S42); a down/up extraction gradient from MapBiomass cropland in gauge buffers, with the placebo re-fit net of a SPEI-by-local-cropland interaction carrying a treated-specific term, since drought-intensified downstream extraction could steepen the downstream slope and mask buffering (Supplementary Tables S43, S44); and a bound on the exposure misclassification implied by both gauge classes sharing the subcuenca-average SPEI (Supplementary Table S45). The paired contrast's own MDE is reported alongside (Supplementary Table S46).

3 Results

3.1 Reservoirs sit where water is already scarce

Reservoirs in Chile are built where drought bites hardest: our 21 dammed basins cluster in the arid centre and north, so a raw comparison would confuse the reservoir with its location. Two comparisons separate them (Figure ??). First, each dammed basin is

compared only with undammed basins sharing its climate, size and elevation (21 of 24
 dammed basins against an effective 91 of 244 controls; Supplementary Figs. S3, S4),
 both evaluated against the drought each experienced, so the contrast is impact per
 unit of deficit. Second, within each dammed basin, regulated reaches below the dam
 are compared against unregulated reaches above it. A genuine reservoir effect must
 appear in these comparisons. Across the fleet none does; the exception, developed
 below, is the carryover-capable minority. The controls are genuinely undammed: only
 10 of the 244 hold a material dam from the 1,370-impoundment national inventory,
 and a re-balanced dam-free control pool leaves every streamflow estimate unchanged
 (Supplementary Table S3).

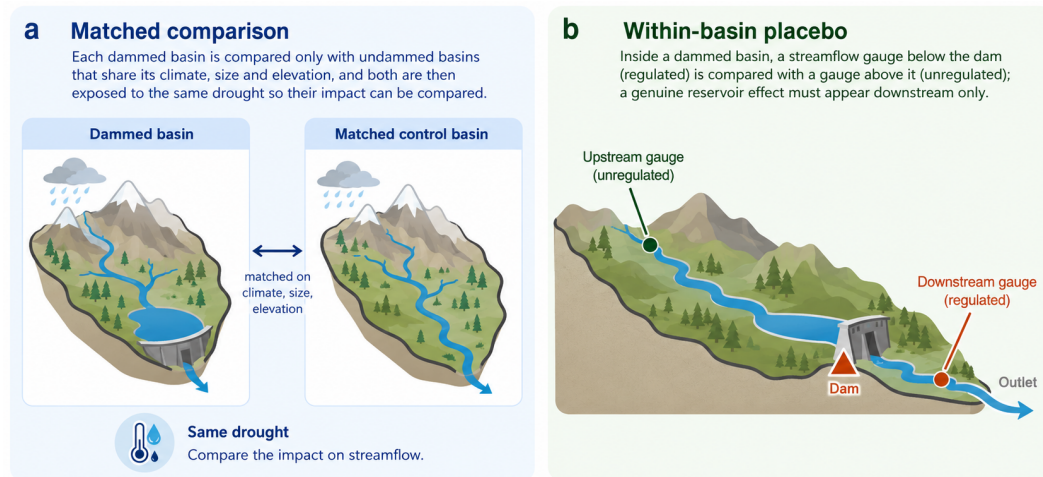


Figure 1: The two comparisons that separate the reservoir from its location. (a) Matched comparison: each dammed basin against undammed basins sharing its climate, size and elevation, both exposed to the same drought. (b) Within-basin placebo: a gauge below the dam against one above it; a genuine reservoir effect must appear downstream only. Study-area map and matching diagnostics: Supplementary Figs. S3, S4.

3.2 Fleet-wide, the apparent buffering is the location, not the dam

The most direct test is on streamflow. Meteorological drought (SPEI-12; the sustained forcing is documented in Supplementary Fig. S5) propagates into streamflow drought (SSI-12) more gently in dammed basins, the visual signature of buffering (Figure ??, panel a). But this flattening tracks where dams sit, not what they do. If the dam

were doing the buffering, transmission would flatten only below it; instead it flattens just as much at the unregulated *upstream* reaches of the same basins (differential slope -0.20 upstream versus -0.16 downstream; Figure ??, panel b). The direct within-basin test confirms the contrast is null: downstream minus upstream is $+0.04$ with a paired permutation $p = 0.72$ (Supplementary Table S29), and a variant giving pseudo-treated controls an elevation-partitioned gauge split leaves the null standard deviation unchanged, so the scheme is not compressing it ($p = 0.73$; Supplementary Table S30). The national gap of -0.18 accordingly collapses to about -0.05 once basins are compared within the same climate region, itself permutation-null ($p = 0.65$; Supplementary Table S28), and in the high-supply south, where buffering should be easiest, the upstream and downstream gaps are identical. The pooled gap does not survive randomization inference ($p_{\text{perm}} = 0.39$), and the decomposition confirms the signal dies only when each climate region carries its own baseline transmission (Supplementary Fig. S6 and Table S47).

Three independent routes through the aridity confound agree. Hard-balancing aridity makes even the *apparent* buffering vanish (intent-to-treat, the all-gauges estimand, -0.02 against -0.18 ; upstream -0.05 against -0.20 ; $p > 0.85$; Supplementary Table S23), a point-estimate shift of about two primary standard errors rather than variance inflation alone. That fit is fragile and we treat it as corroboration only: at an effective control sample of 19 its MDE is near 77% of baseline, its weights are concentrated, and while re-balanced leave-one-out keeps the estimate near zero for seven of the eight top-weighted controls, dropping the most-weighted basin returns -0.184 with balance restored (Supplementary Table S24). No variant turns the null into detected buffering, since every value lies inside the primary permutation null. Second, transmission is genuinely non-linear in aridity (quadratic aridity-by-forcing term $p < 10^{-5}$), and admitting that curvature collapses the apparent intent-to-treat buffering to -0.06 ($p = 0.43$; Supplementary Table S22). Third, the within-region comparison needs no aridity balance at all. The placebo and the within-region estimator therefore carry the

result; the hard balance and the parametric adjustment, which interpolates within support (Supplementary Fig. S2), corroborate it.

Neither the index nor the specification manufactures the null. On log flow the gaps stay null ($p \geq 0.42$) and downstream relative variability is uncompressed (Supplementary Table S12). Re-standardizing SSI on the 1990–2020 baseline leaves every estimate essentially unchanged (contrast $+0.02$, $p = 0.89$; Supplementary Table S11), so the megadrought-heavy reference window is not distorting the slopes, and replacing the year fixed effects with a cubic national time trend, retaining the temporal drought signal they absorb, reproduces every slope (all permutation-null; Supplementary Table S25). Missing streamflow is drought-selective (34% of drought gauge-months fail the completeness rules against 14% otherwise), but the truncation is shared by treated and control gauges and nearly identical across gauge classes (33% downstream versus 35% upstream), so it thins the dry tail for every slope without manufacturing the contrast (Supplementary Table S10). Nor does the pooled slope hide early buffering: at drought onset (2010–2014, storage near-full) the upstream and downstream gaps are identical (-0.43 both), and neither phase shift nor its contrast is significant under permutation (Supplementary Tables S41, S42). A dry-tail effect is likewise absent: the quadratic SPEI interaction is null ($p = 0.65$) and drought-month refits leave every estimate and the contrast null (contrast $p = 0.49$; Supplementary Tables S39, S40).

The placebo’s comparability assumptions survive testing. Upstream gauges sit 0.54 km higher, and elevation genuinely flattens transmission among control gauges (-0.21 per km, $p < 0.001$, $n = 84$, from a positive baseline of 0.53; Supplementary Table S31), but the gradient works *against* the finding: it alone would make upstream look more buffered, and the gap times that sensitivity over-explains the observed -0.04 difference, leaving at most 0.07 of downstream-specific buffering, 13% of baseline (Supplementary Table S21). Cumulative scale behaves the same way, flattening transmission among controls ($+0.08$ per log discharge, $p < 0.001$) while the paired

basins differ little in scale. Fitting the placebo net of each gradient leaves every effect null and the contrast unchanged (elevation-adjusted $D = +0.02$; discharge-adjusted $D = -0.03$; Supplementary Tables S33, S34). Local relief, separating bedrock from alluvial reaches, likewise leaves it unchanged, moving the downstream estimate toward the upstream value rather than away (Supplementary Table S35); the snowmelt confound is excluded on the differential estimand across four specifications; and the exposure error implied by the shared subcuenca forcing is about 0.01 units, asymmetric in the direction that works against downstream-specific buffering (Supplementary Table S45).

Gauge classification survives three validations. Against the DGA hydrographic hierarchy, 67 of 70 downstream gauges lie in a sub-cuenca named for the dam's own river; excluding the three exceptions leaves the contrast null ($D = +0.04$), as does excluding the sixteen gauges within 100 m of the dam ($D = +0.12$), both moving the downstream estimate toward zero rather than toward buffering (Supplementary Table S36). The mirror-image concern is cascade contamination, an upstream gauge itself below *another* dam, whose flattened transmission would be real regulation misread as siting. Screening all 35 upstream gauges against the national inventory, the placebo survives every tier, from excluding the three flagged within their own sub-watershed ($D = +0.08$) to the conservative cuenca-wide bound stripping 20 of 35 gauges ($D = +0.05$), every contrast permutation-null (Supplementary Table S38). The direction of change is diagnostic: were the upstream flattening hidden regulation, removing flagged gauges should steepen the placebo toward zero, and instead it deepens. Snapping every gauge and dam to the HydroRIVERS network closes the twin concern that elevation-classified upstream gauges sit on parallel, disconnected tributaries: the strictly network-validated same-channel contrast is *positive* ($D = +0.12$, $p = 0.18$), the unregulated reaches more attenuated than the regulated ones (Supplementary Table S37). The contrast's power is disclosed rather than assumed: with 15 paired basins it detects only gaps above 0.28 SSI units,

about 52% of baseline, so moderate downstream-specific buffering is not excluded (Supplementary Table S46).

A skeptical reading asks whether the flat contrast hides a real buffer masked by the downstream extraction gradient. The gradient is real, cropland within 5 km of a downstream gauge averaging 0.20 of the buffer area against 0.08 upstream, but it cannot hide a buffer: local extraction does not predict the transmission slope among control gauges ($p \geq 0.41$), its point estimate carries the sign opposite the masking mechanism, and re-fitting the placebo net of a SPEI-by-local-cropland interaction leaves the contrast null (-0.03 , $p = 0.83$; Supplementary Tables S43, S44).

3.3 The powered nulls exclude operational buffering; demand pathways are bounded

The same answer holds across the remaining pathways (Table ??): every estimate that carries evidential weight brackets zero: the cropland-area expansion ATT (-0.69 ha km⁻², 95% confidence interval $[-2.4, 1.1]$) and its forcing-conditioned slope gap ($p_{\text{perm}} = 0.91$). That slope gap carries a structural caveat: central Chile’s cropland is dominated by perennial orchards and vineyards, which cannot flex with annual SPEI, so a null interaction is expected on rigidity grounds alone, and the informative cropland test is the dynamic event study below. The evapotranspiration pathway is excluded rather than counted as a null: whole-basin ET is confounded by barren-land aridity with failed pre-trends, and the post-hoc orchard stratum relies on MODIS MOD16, whose documented bias toward zero over irrigated land, a product property identified independently of our results, disqualifies its null (Supplementary Table S16, Fig. S7).

The null is informative but its power is concentrated in streamflow. With about 21 dammed clusters the design cannot claim exact equivalence to zero but rules out operational buffering above 46 to 59% of baseline, and a positive control confirms it: injected buffering is recovered one-to-one and detected beyond that threshold

Apparent streamflow buffering is siting, not the reservoir

Effect is permutation-null and as strong in UPSTREAM (unregulated) gauges

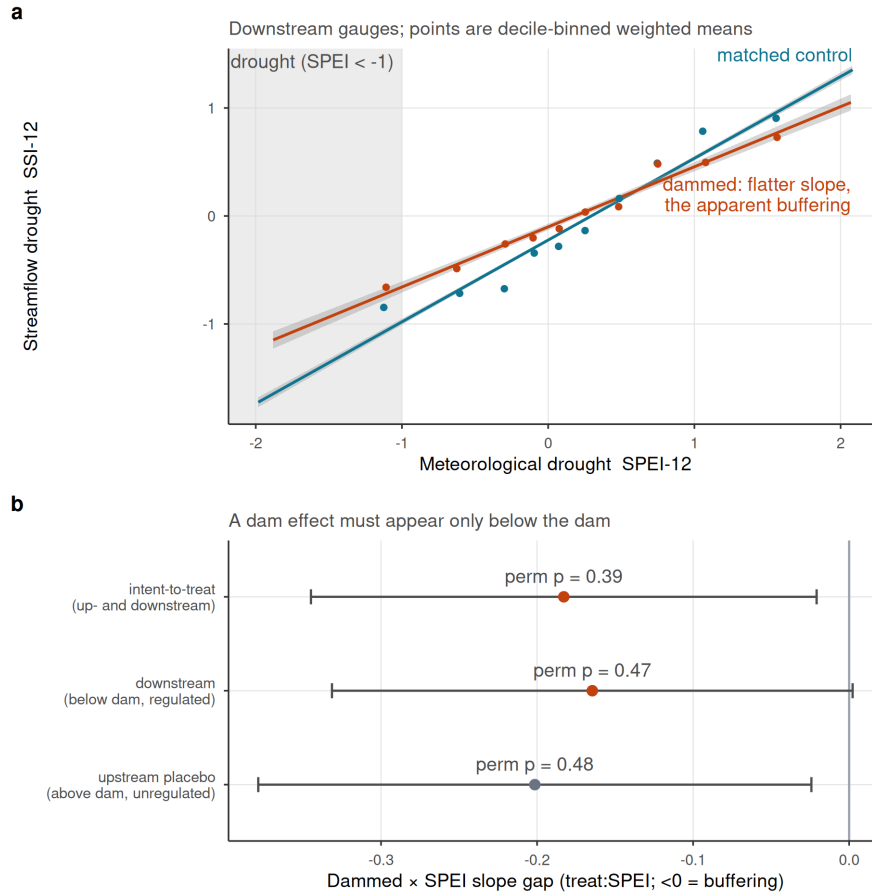


Figure 2: Fleet-wide, streamflow buffering is reservoir siting, not regulation. (a) SPEI-12 to SSI-12 transmission for dammed versus matched-control basins (downstream gauges); weighted least-squares fits with 95% bands. (b) Differential slope (treat:SPEI; <0 = buffering) for intent-to-treat, downstream-only, and the upstream-only placebo, with randomization-inference p-values. All are permutation-null and the upstream placebo is as strong as downstream, so the attenuation is siting, not the dam.

(Supplementary Fig. S8 and Table S48). This is a wide null: a 30 to 40% reduction in transmission is fully compatible with the data, so the finding is no *large* operational buffering, not none. It also covers a record ending in March 2020, spanning the first decade of the megadrought but not the 2019–2022 peak deficits, which the storage and cropland pathways carry to 2024. Only streamflow excludes material effects at operationally relevant magnitudes: the cropland and evapotranspiration designs have MDEs far above baseline (7,330% and 237%; Supplementary Table S49), so their nulls corroborate rather than independently establish absence.

Pooling the fleet, moreover, hides a heterogeneity by storage capacity. We label this split exploratory at the outset: it is a subgroup analysis of seven treated units inside a large robustness battery, and no multiplicity adjustment is applied, so we read it as hypothesis-generating. Split by the carryover ratio on the buffering sample of 17 basins, the seven carryover reservoirs show a downstream-specific buffering the seasonal ones do not: slope -0.46 downstream versus -0.23 upstream, a contrast of -0.23 ($p = 0.014$; Supplementary Table S50), essentially unchanged on the six strictly paired basins ($D = -0.234$, $p = 0.004$; Supplementary Table S8) and surviving the elevation-partitioned null ($p = 0.025$; Supplementary Table S30); the ten seasonal reservoirs show no such contrast ($D = +0.18$, $p = 0.26$), their estimate being if anything positive, the direction drought-intensified diversion would produce though indistinguishable from zero. The signal’s fragility is statistical rather than mechanistic. It is not a hydropower-scheduling artifact, since excluding the two dams operated also for generation *strengthens* the estimate on the five irrigation-only carryover basins ($D = -0.31$, $p = 0.057$); but the contrast, while negative under every leave-one-out, carries a permutation p ranging from 0.004 to 0.12, and recomputing the ratio with unregulated upstream-gauge flow moves one seasonal basin into the group, under which split it is still negative but no longer significant ($D = -0.165$, $p = 0.146$; Supplementary Tables S7, S9). The direction is robust to the denominator and to single-basin exclusion; the significance is robust to neither. Within those bounds,

Table 1: Reservoir-effect estimates across estimators: cross-sectional doubly-robust matched ATTs (including the water-rights induced-demand test) and forcing-interacted difference-in-differences (DiD) slope gaps (treat:SPEI). The DiD row and water-rights accrual are judged by randomization-inference p-values, the design’s primary inference at ~21 clusters; cluster-robust confidence intervals over-reject here and are shown for completeness. A positive estimate would indicate induced-demand vulnerability; a negative one, operational buffering.

Group	Outcome	Estimand	Estimate	95% CI	p	Verdict
Matched ATT	Cropland-area expansion	ha km ⁻² added	-0.687	[-2.44, 1.06]	-	null
Matched ATT	Water-rights accrual	rights / 100 km ² added	20.036	[4.02, 36]	0.52	null
Forcing-interacted DiD	Cropland area	differential deficit-to-impact slope	0.001	[-0.00344, 0.00493]	0.91	null

reservoirs with enough storage to carry water across years appear to buffer multi-year drought while seasonal ones, which physically cannot, show nothing, so the aggregate null is driven by the seasonal majority.

3.4 Reservoirs mark prior vulnerability; megadrought divergence is bounded, not absent

Dammed basins carry the vulnerability reservoirs are blamed for concentrating, about 4.5 times more cropland than matched controls (entropy-balancing-weighted means 10.8% versus 2.4% of basin area, with 95% bands in Figure ??, panel a), but that gap is a *siting* signature stable across the record. A dynamic event study finds the differential trend flat before 2010 (cluster-robust $p = 0.65$, permutation $p = 0.72$) with no post-2010 divergence detected ($p = 0.81$; Figure ??, panel b), and the null is metric-invariant, likewise permutation-null on log cropland share ($p = 0.37$). Both are bounded nulls, not confirmations of parallelism: the pre-trend test detects only divergence accumulating to about half the control cropland level by 2009, and the permutation envelope widens after 2015 (Supplementary Table S51): reservoirs mark where cropland already was, and within these bounds the record is consistent with no megadrought-era expansion, though late-panel divergence remains possible.

The same holds for the most direct demand measure. In the DGA registry of con-
 sumptive water rights, dammed basins appear to accrue rights faster than controls,
 on both the outlier-robust count (+20 per 100 km²; Table ??, Supplementary Fig.
 S9) and the winsorized volume (+2.5 l/s per km²), each with an analytic interval
 excluding zero. This is the naive-analysis trap the design catches, since arid basins
 carry denser water-rights activity. The count does not survive randomization inference
 ($p_{\text{perm}} = 0.52$) and remains null under a fully non-parametric match on the aridity-
 overlap subset ($p_{\text{perm}} = 0.42$; Supplementary Table S20). The winsorized volume is
 more delicate: insensitive to the winsorization threshold on the full sample, it reaches
 $p = 0.005$ on the smaller aridity-overlap subset under unit-level permutation, yet
 those residuals are spatially autocorrelated and the effect is not significant under
 the governing cuenca-block permutation ($p_{\text{perm}} = 0.077$), the verdict the ranked
 raw volumes also give. The water-rights evidence is therefore not uniformly null:
 the count is cleanly so, the ranked volumes are null under spatial inference, and the
 winsorized volume sits closer to the boundary, flagging a spatial concentration of
 large consumptive rights in dammed cuencas that the design can bound but cannot
 attribute to siting versus induced demand at this grain, and that warrants parcel-level
 investigation. Two institutional statements must be kept apart. Administrative
 closure did not mechanically silence the registry differentially: treated basins entered
 the drought with a higher allocated stock yet accrued new rights about 2.7 times
 faster than their weighted controls, and none went without new grants. But closure
 does make new grants a weak operationalization of total demand, because in closed
 basins rights change hands through private market transfers recorded outside the
 registry, so market-driven reallocation is invisible to the accrual measure; cropland
 area is the physical check against concentration at scale, and a transaction-level test
 would be needed to close the pathway.

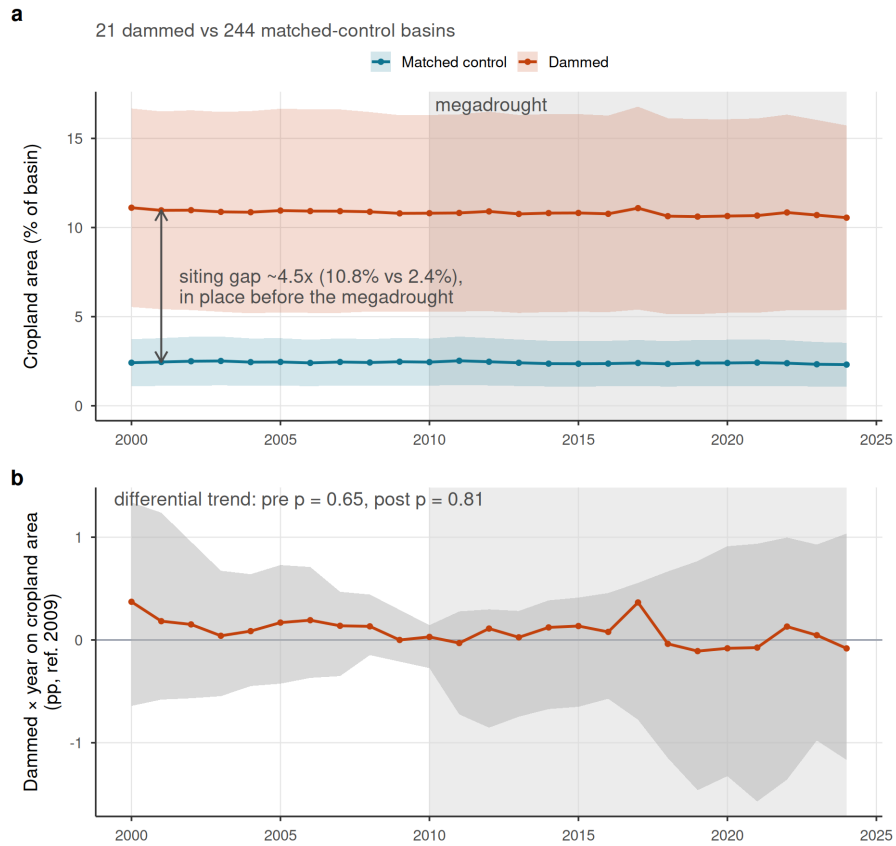
Reservoirs do not alter irrigated-area dynamics

Figure 3: Cropland-area dynamics in dammed versus matched-control basins (21 dammed, 244 controls). (a) Entropy-balancing-weighted mean cropland area (% of basin, Map-Biomass class 18) with 95% bands, y-axis anchored at zero: dammed basins hold ~4.5x more cropland but the trajectories evolve in parallel. (b) Dynamic event study (dammed x year, reference 2009); the grey band is the 95% randomization-inference null envelope, cluster-robust intervals being anti-conservative at ~21 treated clusters. The envelope widens after 2015, so the flat late trajectory is a bounded null. The megadrought is shaded.

3.5 Storage falls; the carryover buffer shrinks

If streamflow excludes material operational buffering and the demand pathways show no amplification within their detection limits, what matters is how much water there is to store (Figure ??). The decline is not a smooth drift: the peak steps down at the 2010 megadrought onset while the trough keeps eroding through it. Because 22 to 26 reservoir clusters are too few for cluster-robust inference, every storage p-value below is a null-imposed wild cluster bootstrap (Supplementary Table S6). Over 2005–2024 peak storage falls by 1.3 percentage points of capacity per year ($p = 0.004$) and the trough by 1.2 pp yr⁻¹ ($p = 0.001$), with no detectable amplitude trend ($p = 0.98$; Supplementary Table S52); nested at the 2010 split, the peak steps down (-0.19 of capacity, $p = 0.009$) then shows no further drift, while the trough, whose step is not significant, keeps falling (-0.010 yr⁻¹, $p = 0.009$; Supplementary Table S53). Bootstrap inference tightens rather than weakens these: the peak rests at a lower equilibrium while the carryover floor still erodes. Nor is the pattern an artifact of equal-weighting a fleet whose capacities span three orders of magnitude: weighted by capacity the trough decline is steeper rather than weaker (-1.7 pp yr⁻¹, $p = 0.001$) and the amplitude widens rather than narrows ($p = 0.031$), so the flat unweighted and widening weighted amplitudes both run opposite to the narrowing that sedimentation or a failing refill cycle would produce. Weighting costs precision, though: the weighted peak trend and both weighted 2010 steps are not significant under the bootstrap, so the step evidence rests on the unweighted fit. In absolute volume the coverage-stable subset (22 reservoirs, 98% of fleet capacity) lost 88 hm³ yr⁻¹ of peak and 209 hm³ yr⁻¹ of trough storage, and restricting the trends to that subset, so mid-panel commissioning cannot tilt them, leaves the step-versus-drift separation unchanged.

The decline directly shrinks the carryover volume that buffers multi-year drought: peak storage fell from 87% of capacity before 2010 to 67% after and the trough from 40% to 25%, so the pool available to buffer a second and third dry year is materially

smaller. Whether the decline is supply-driven or management-driven cannot be attributed causally, but it is directionally consistent with a supply-side reading resting on an independent decline: unregulated control streamflow fell about 2.9% per year over 2005–2020 (dammed 3.4%), larger than the 1.3 pp yr⁻¹ fall in the storage stock, as expected since storage partially absorbs inflow shocks, and the megadrought runoff deficit (Alvarez-Garreton et al., 2021; R. D. Garreaud et al., 2017) is larger still in percentage terms. Sedimentation is a third candidate the percent-of-capacity metric cannot exclude by construction, since registry capacities are static nominal values and silting of active storage in these high-sediment catchments would mechanically depress the reported fraction at constant inflow. Three features bound that reading: sediment accumulation is gradual, whereas the peak decline is a level step; a large loss of live volume would compress the seasonal amplitude, which does not narrow; and a sediment-driven decline would not co-move with the independent streamflow fall, which it does. A sediment contribution to the continuing trough erosion cannot be excluded without resurveyed bathymetry, which we flag as the direct test. Storage is a mass balance, so the band alone cannot separate falling inflows from sustained releases; our direct demand outcomes point the same way, with neither cropland area nor water-rights allocation showing expansion (Figure ??, Table ??) (Li et al., 2023; Wang et al., 2024).

4 Discussion

Across a matched national design, multiple estimators, and a within-basin falsification test, the apparent drought buffering attributed to reservoirs is, for the seasonal majority of the fleet, a measurable siting confound. The upstream placebo is the decisive evidence: a regulation effect must appear only below the dam, and across the fleet it does not; what remains in the aggregate is the aridity gradient. The exception is itself informative: the seven carryover-capable reservoirs show a downstream-specific contrast exactly where a real regulation effect should sit, an exploratory signal bounded

Storage declines; a change in refill amplitude cannot be resolved

The whole band drifts down; the peak-trough amplitude trend is inconclusive (wide interval)

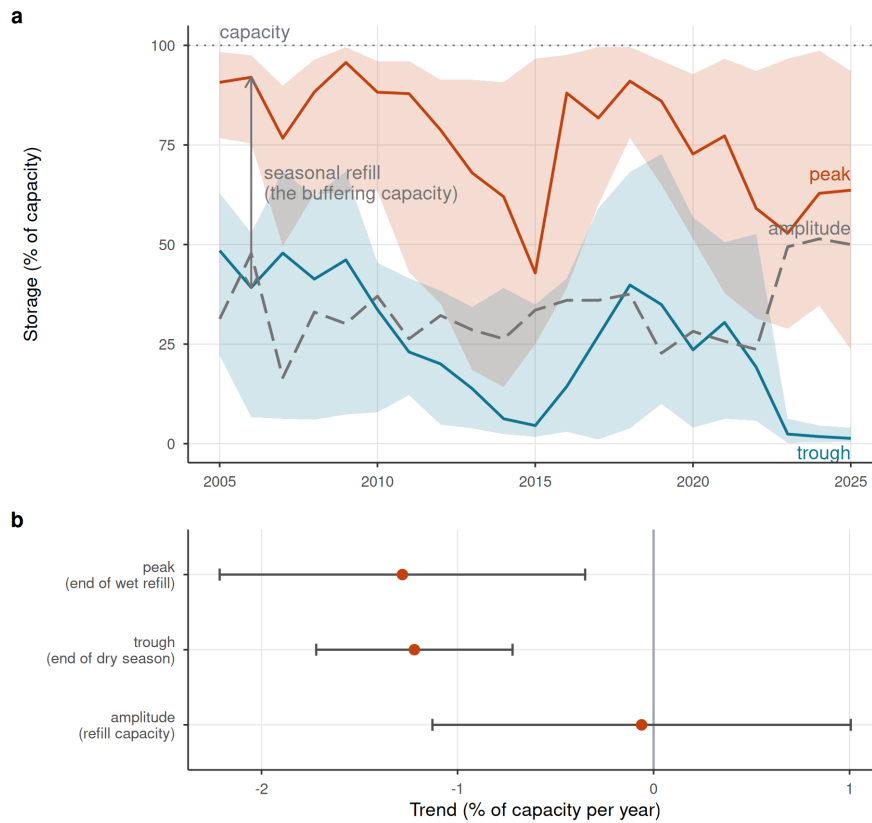


Figure 4: Reservoir storage declines; the carryover buffer shrinks with supply. (a) Cross-reservoir annual storage band: realized median peak and trough percent-of-capacity (solid) with interquartile ribbons, and the peak-trough amplitude (grey dashed). Medians are used because a few reservoirs report storage above nominal capacity (dotted 100% line); the band drifts down while the amplitude does not narrow. (b) Per-year trend slope of peak, trough, and amplitude (reservoir fixed effects), with 95% cluster-robust intervals; every coefficient is confirmed by wild cluster bootstrap (Supplementary Table S6). Descriptive: pooled within-reservoir trend, no control group.

below, so the null is a statement about storage-limited reservoirs, not about storage as such. Dammed basins hold about 4.5 times more cropland than matched controls, but that gap is fixed across the record, with parallel pre-trends and no megadrought divergence detected within the design’s power (a bounded null, not a confirmed absence). One reading fits: reservoirs mark where water-dependent vulnerability already concentrated rather than make it. This is a finding, not a failure of power: the apparent effect has an identified cause, and the positive control shows the design detects buffering when it is present.

The corollary is transferable: dammed-versus-undammed or before-after contrasts inherit the full siting confound (Lee et al., 2026; López-Moreno et al., 2009; Schilstra et al., 2024; Wu et al., 2017).

If the streamflow pathway excludes material operational buffering at multi-year scale and the demand-side outcomes are consistent with no amplification at their respective detection limits, what matters is the water available to store. The storage band steps down at the megadrought onset, peak and trough together, which directly shrinks the carryover volume held across years, the buffering mechanism for multi-year drought: the pool available to carry a second and third dry year is materially smaller by 2024 than it was in 2010. This is a real reduction in multi-year buffering capacity. Whether it is supply-driven or management-driven cannot be attributed causally, since the trend is descriptive and confounded with operational change; it is directionally consistent with a supply-side reading. The seasonal amplitude is unchanged within a wide interval, indicating that the reservoirs still refill to their seasonal pattern when water arrives, consistent with the carryover loss tracking falling inflow rather than a failure of the refill cycle. The powered result is the level decline, the expected mass-balance outcome of persistent inflow deficits; descriptive and confounded with management change, it supports no causal attribution but is directionally consistent with the independent decline in unregulated control streamflow, which fell faster (2.9% per

year) than the storage stock (1.3 percentage points of capacity per year), as a buffer should partially absorb an inflow shock. Read against Chile’s structural aridification (Alvarez-Garretton et al., 2021; R. D. Garreaud et al., 2017; Zambrano et al., 2025), accelerating Andean warming and snow loss now cutting into the snow-fed streamflow these basins depend on (Kim et al., 2025), and the global pattern of reservoirs failing to refill (Li et al., 2023; Wang et al., 2024), the binding constraint is shifting toward supply.

These findings give the contested socio-hydrological *reservoir effect* and *supply-demand cycle* (Di Baldassarre et al., 2018; Kellner, 2021) a credibly identified test, and find the protective function over-attributed to storage. Because treatment is time-invariant, we test the spatial imprint of these feedbacks, not the longitudinal loop. The 4.5-fold cropland gap could be the equilibrated legacy of demand induced at construction, and the record opens in the scarcity phase, after any expansion under perceived abundance, so our null bounds drought-era expansion, not the cycle as a whole. Within scope the nulls refine the theory: the cycle leaves no operational imprint during drought, so its mechanisms likely act through siting and construction rather than operation, a distinction current models omit. How far the demand-side null travels depends on governance: the siting lesson holds wherever dams are placed non-randomly; prior appropriation, which forfeits unused rights, may amplify induced demand while adaptive allocation curtails it. Whether the null is general or the mark of a fully allocated market we cannot adjudicate: rights kept accruing in dammed basins, but systems with water still unallocated could express an induced demand that ours cannot. The upstream placebo itself ports more broadly. Short, steep catchments with gauges above and below dams are common across the world’s regulated mountain rivers, from the Sierra Nevada to the Pyrenees, the Alps, and the Andean cordillera beyond Chile. In large low-gradient basins such as the Murray-Darling or the Indus plains, floodplain storage and transmission losses would blur the above/below contrast and routing-corrected lags or naturalized-flow counterfactuals would be needed. The

method is therefore not universal, but its scope conditions are explicit and the paired gauges it requires exist across much of the world’s irrigated mountain-front hydrology.

Chile’s 2022 Water Code reform moves toward the adaptive allocation our results argue for (Macpherson et al., 2023) but postdates our window. Under structural aridification, building more storage is a supply-blind response to a supply problem: the buffering dams provide is a property of where they sit, not of what they do (Ho et al., 2017). Adaptation must shift toward demand management, reallocation and managed aquifer recharge (Grafton et al., 2018; Rivera-Vidal et al., 2025; Scanlon et al., 2023).

Several limitations bound these conclusions, each quantified in the Supplementary Information. The result is a bounded null: the design excludes streamflow buffering above 46 to 59% of baseline, demand-side outcomes are far less powered and the pre-trend test power-limited (Supplementary Tables S49, S51), so causal weight rests on the placebo, demand nulls corroborating; all survive the hard-balanced match, itself a fragile corroboration at its small effective sample, whose weight concentration and leave-one-out sensitivity are disclosed (Supplementary Tables S23, S24). The powered streamflow null also ends in March 2020, before the most acute 2019–2022 deficits, so the late-phase storage and cropland pathways carry that part of the record. The early-to-late phase analysis is likewise null under the design’s permutation inference, with neither the downstream nor the upstream phase shift significant and their contrast far from it (Supplementary Table S42), so the flat late-phase downstream estimate is a noisy (standard error 0.36) point within its null rather than evidence of regulation; cluster-robust intervals, which the design’s own rule says over-reject at about 21 clusters, overstate the phase signal. The registry measures paper rights, not wet water: Chilean basins are widely over-allocated and granted rights can sit unused or hoarded (Barría et al., 2021; Reade Malagueño & D’Odorico, 2024), so under drought, when physical availability binds, the accrual we test is an

entitlement response that non-use could dampen; cropland is the physical check. The registry also records original grant dates, not the market transfers through which rights change hands in administratively closed basins (Reade Malagueño & D’Odorico, 2024), so concentration below reservoirs would be invisible to the accrual measure; cropland trajectories argue against it at scale, but a transaction-level test would be needed to close this pathway. Both metrics also miss intensification within an unchanged footprint; central Chile’s documented shift toward thirsty export fruit (Reade Malagueño & D’Odorico, 2024) is exactly such a transition, invisible to both; a consumption-grade test awaits thermal ET or crop-transition maps. Groundwater substitution is bounded, dammed and control water tables falling at indistinguishable rates across 213 DGA wells despite severe regional drawdown (Taucare et al., 2024); monitored wells need not sample the orchard footprints where pumping concentrates, so unmonitored drawdown could sustain a historically reservoir-induced footprint (Supplementary Table S17). In that reading the reservoir effect is not absent but displaced to the aquifer, a construction-era legacy persisting on a pool the design cannot see; the well bound and flat cropland trajectories argue against substitution at scale but do not close the pathway. The snowmelt confound, acute in these snow-fed Andean basins (Ayala et al., 2025; Götte & Brunner, 2024), is tested on the differential estimand and excluded across four specifications (Supplementary Table S15). The geomorphic check leaves the placebo unchanged, though 2.5 km relief cannot resolve aquifer storage itself, a residual caveat (Supplementary Table S35). Cascade contamination of the upstream placebo, upstream gauges that are themselves regulated by another dam, is screened against the 1,370-dam national inventory and excluded at three tiers, and removing the flagged gauges deepens rather than removes the upstream attenuation (Supplementary Table S38); the classification is likewise validated against routed network topology, where the strictly same-channel contrast turns positive, unregulated reaches more attenuated than regulated ones (Supplementary Table S37). Streamflow missingness is drought-selective but common

across gauge classes, so it truncates the dry tail without generating the null contrast (Supplementary Table S10). The placebo also spans land-cover and abstraction contrasts: within paired basins, downstream valleys carry about two and a half times the cropland of upstream headwaters (per-basin means, 12 of 15 basins downstream-richer; Supplementary Table S43), and upstream reaches carry unrecorded diversions. Gauge fixed effects absorb these level differences, and the gradient cannot mask a buffer: local cropland intensity does not predict the transmission slope among control gauges ($p = 0.41$), the point estimate carries the sign opposite to the masking mechanism, and re-fitting the placebo net of a SPEI-by-local-cropland interaction leaves the contrast null (-0.03 , $p = 0.83$; Supplementary Tables S43, S44). The cropland sensitivity is identified among control gauges, so a release-sustained extraction response could exceed it; the adjusted model therefore lets treated reaches carry their own SPEI-by-cropland term, and the implied masking magnitude sits two orders below the contrast MDE either way. Administrative closure cannot force the demand null, since treated basins kept accruing rights faster than their controls (Supplementary Table S2); what it does do is make grant accrual a weak proxy for total demand. No matched pair shares a watershed; outbound transfers would shrink the gap and imports would mimic downstream-specific buffering, which is absent (Supplementary Table S2); irrigation-only restriction leaves the result unchanged (Supplementary Table S1). MOD16's bias toward the null excludes the ET pathway from evidence (Supplementary Table S16); the demand conclusion rests on cropland and the registry, with thermal ET as future work. The 12-month accumulation targets multi-year vulnerability, not the sub-seasonal buffering reservoirs also provide, which we neither test nor dispute. The fleet is partly seasonal by design: the median treated reservoir stores 0.3 years of downstream flow, and only seven of seventeen hold at least half a year (Supplementary Table S54). This heterogeneity is informative: the seven carryover-capable reservoirs show a downstream-specific buffering the seasonal ones do not (contrast -0.23 , $p = 0.014$; Supplementary Table S50), so the aggregate null

is a statement about the seasonal majority, not about reservoirs that can actually carry water across years. Reservoirs with enough storage to bridge a dry year appear to buffer multi-year drought; seasonal ones, which physically cannot, do not. The signal survives the confounds we can test but not every inferential stress. It is not hydropower scheduling: excluding the two dams operated also for generation (Colbún, Lago Laja) strengthens the contrast on the five irrigation-only carryover basins ($D = -0.31$, $p = 0.057$; Supplementary Table S9). It is directionally stable under every leave-one-out, but its permutation p spans 0.004 to 0.12 across single-basin exclusions (Supplementary Table S9), and the threshold is classification-bounded: recomputing the carryover ratio with unregulated upstream-gauge flow, the available inflow proxy, keeps every carryover basin classified as carryover but reclassifies one seasonal basin into the group, under which split the carryover contrast is still negative but no longer significant ($D = -0.165$, $p = 0.146$; Supplementary Table S7). Together with the absence of any multiplicity adjustment across the paper’s robustness battery, this is why the split is labeled exploratory. It bounds the generalization: the headline null applies to the storage-limited fleet, and the capacity-threshold signal is a directionally robust candidate for a rule-curve or commissioning-based test with more carryover reservoirs. Two extensions can test what our window cannot: a staggered event study of cropland around historical commissioning would probe construction-phase induced demand; a rule-curve quasi-experiment on release records, once public, would isolate operation.

5 Conclusions

At the multi-year timescale of structural aridification, then, the design detects no operational buffering above 46 to 59% of baseline across the fleet and no megadrought-era demand expansion beyond the siting legacy: for the seasonal majority, the apparent buffering is a statistical shadow of where dams sit, not of what they do, while the water available to store declines. The qualification is physical: the seven

carryover-capable reservoirs, holding at least half a year of downstream flow, do show downstream-specific buffering, an exploratory but directionally robust signal, so the aggregate null describes the seasonal majority. The nulls are bounds, not zeros; effects inside them can be volumetrically material (Supplementary Table S26). But under structural aridification, with no operational buffering within the design’s bounds for the storage-limited fleet, building more storage is a supply-blind response to a supply problem. Chile continues to plan new reservoir construction to expand storage, while its agricultural water demand continues to grow under perpetual private water rights that the 2022 Water Code reform only begins to address (Macpherson et al., 2023). The alternative is not theoretical: demand reallocation through active water markets, irrigation efficiency, managed aquifer recharge, and crop transitions toward lower water footprints are available now and do not depend on rainfall (Grafton et al., 2018; Rivera-Vidal et al., 2025; Scanlon et al., 2023). Resilience to a falling supply ceiling will come from confronting demand and allocation, not from more impoundments.

Acknowledgments

The National Research and Development Agency of Chile (ANID) funded this study through the drought emergency project FSEQ210022 and Fondecyt Iniciación N°11190360.

Open Research

The full analysis pipeline, comprising the `targets` workflow, the reusable R functions (`src/R/`), the study-parameter configuration (`config/`), and the scripts that generate every figure and table, is openly available at https://github.com/frzambra/dam_drought_effectiveness and archived on Zenodo (Zambrano Bigiarini, 2026). That archive also holds the reservoir storage records and the national dam registry, the derived result tables, and the matched-set weights and metadata, and is the citable version of record for both the software and the data generated here. All analyses run in R, using `WeightIt/cobalt` for balancing and diagnostics, `fixest`

for fixed-effects regression, and `sf/terra/exactextractr` for the spatial extractions;
 the pinned R environment (`renv.lock`) reproduces the package versions used. Every
 result in the manuscript corresponds to a named pipeline target, so the analyses can
 be regenerated with a single `targets::tar_make()`.

The third-party input datasets are publicly available from their providers:

- **Precipitation and temperature.** Climate Hazards Group Infrared Precipitation with Stations (CHIRPS) and Climate Hazards Center Infrared Temperature with Stations (CHIRTS), Climate Hazards Center, University of California Santa Barbara, <https://www.chc.ucsb.edu/data>. The SPEI-12 forcing and the P/PET aridity index were derived from these products for this study on a 1991–2020 reference period. The national index rasters are too large to archive with the repository; the pipeline extracts the basin-level series from them, and that extraction code and the resulting analysis tables are in the archive.
- **Evapotranspiration.** MODIS MOD16A2GF version 061 gap-filled 8-day actual evapotranspiration at ~500 m, NASA Land Processes Distributed Active Archive Center, <https://doi.org/10.5067/MODIS/MOD16A2GF.061>.
- **Land cover.** MapBiomas Chile Collection 2, annual 30 m land-cover and land-use classification, <https://chile.mapbiomas.org>.
- **Streamflow.** Daily mean streamflow for 811 DGA gauges, February 1913 to March 2020, compiled by the Center for Climate and Resilience Research (CR2), <https://www.cr2.cl>, distributed as `cr2_qflxDaily_2020`.
- **Snow water equivalent.** ERA5-Land monthly averaged reanalysis, snow depth water equivalent (variable `sd`) at 0.1°, Copernicus Climate Change Service Climate Data Store, catalogue entry `reanalysis-era5-land-monthly-means`, documented by Muñoz-Sabater et al. (2021).

- 848 • **Elevation.** SRTM 3 arc-second (~90 m) void-filled digital elevation model,
 849 CGIAR-CSI SRTM 90 m Digital Elevation Database, <https://srtm.csi>
 850 [.cgiar.org](https://srtm.csi.cgiar.org).
- 851 • **Climate classification.** Köppen-Geiger classes for the 1991–2020 normal
 852 period at ~1 km (Beck et al., 2023).
- 853 • **River network.** HydroRIVERS version 1.0, used for the routed-topology vali-
 854 dation of gauge classification, HydroSHEDS, [https://www.hydrosheds.org/](https://www.hydrosheds.org/products/hydrorivers)
 855 [products/hydrorivers](https://www.hydrosheds.org/products/hydrorivers).
- 856 • **Orchard cadastre.** Catastro Frutícola, Centro de Información de Recursos
 857 Naturales (CIREN) and Oficina de Estudios y Políticas Agrarias (ODEPA),
 858 <https://www.odepa.gob.cl>.
- 859 • **Reservoirs, watersheds, water rights, and wells.** Reservoir storage
 860 records, the national dam registry (Embalse_2026_05_31), the Banco Nacional
 861 de Aguas cuenca and subcuenca hydrographic boundaries, the consumptive
 862 water-rights registry, and well hydrographs, all from the Dirección General de
 863 Aguas, Ministerio de Obras Públicas, Chile, <https://dga.mop.gob.cl>.

864 References

- 865 Alvarez-Garreton, C., Boisier, J. P., Garreaud, R., Seibert, J., & Vis, M. (2021).
 866 Progressive water deficits during multiyear droughts in basins with long hydro-
 867 logical memory in Chile. *Hydrology and Earth System Sciences*, 25(1), 429–446.
 868 <https://doi.org/10.5194/hess-25-429-2021>
- 869 Ayala, Á., Muñoz-Castro, E., Farinotti, D., Farías-Barahona, D., Mendoza, P. A.,
 870 MacDonell, S., et al. (2025). Less water from glaciers during future megadroughts
 871 in the Southern Andes. *Communications Earth & Environment*. [https://](https://doi.org/10.1038/s43247-025-02845-6)
 872 doi.org/10.1038/s43247-025-02845-6
- 873 Barriá, P., Sandoval, I. B., Guzman, C., Chadwick, C., Alvarez-Garreton, C., Díaz-
 874 Vasconcellos, R., et al. (2021). Water allocation under climate change. *Ele-*

- 875 *menta: Science of the Anthropocene*, 9(1), 00131. [https://doi.org/10.1525/](https://doi.org/10.1525/elementa.2020.00131)
876 [elementa.2020.00131](https://doi.org/10.1525/elementa.2020.00131)
- 877 Beck, H. E., McVicar, T. R., Vergopolan, N., Berg, A., Lutsko, N. J., Dufour, A., et
878 al. (2023). High-resolution (1 km) Köppen-Geiger maps for 1901–2099 based on
879 constrained CMIP6 projections. *Scientific Data*, 10, 724.
- 880 Boisier, J. P., Alvarez-Garreton, C., Cordero, R. R., Damiani, A., Gallardo, L.,
881 Garreaud, R. D., et al. (2018). Anthropogenic drying in central-southern Chile
882 evidenced by long-term observations and climate model simulations. *Elementa*,
883 6(1), 74. <https://doi.org/10.1525/elementa.328>
- 884 Chalise, D. R., Sankarasubramanian, A., & Ruhi, A. (2021). Dams and climate
885 interact to alter river flow regimes across the united states. *Earth's Future*, 9(4).
886 <https://doi.org/10.1029/2020EF001816>
- 887 Chalise, D. R., Sankarasubramanian, A., & Ruhi, A. (2023). Spectral signatures of
888 flow regime alteration by dams across the united states. *Earth's Future*, 11(2).
889 <https://doi.org/10.1029/2022EF003196>
- 890 Di Baldassarre, G., Wanders, N., AghaKouchak, A., Kuil, L., Rangelcroft, S.,
891 Veldkamp, T. I. E., et al. (2018). Water shortages worsened by reservoir
892 effects. *Nature Sustainability*, 1(11), 617–622. [https://doi.org/10.1038/](https://doi.org/10.1038/s41893-018-0159-0)
893 [s41893-018-0159-0](https://doi.org/10.1038/s41893-018-0159-0)
- 894 Garreaud, R., Boisier, J. P., Álvarez-Garreton, C., Christie, D., Carrasco-Escaff,
895 T., Vergara, I., et al. (2025, March). Hyperdroughts in central Chile: Drivers,
896 Impacts and Projections. Hydrometeorology/Instruments; observation techniques.
897 <https://doi.org/10.5194/egusphere-2025-517>
- 898 Garreaud, R. D., Alvarez-Garreton, C., Barichivich, J., Boisier, J. P., Christie, D.,
899 Galleguillos, M., et al. (2017). The 2010–2015 megadrought in central Chile:
900 Impacts on regional hydroclimate and vegetation. *Hydrology and Earth System*
901 *Sciences*, 21(12), 6307–6327. <https://doi.org/10.5194/hess-21-6307-2017>

- Götte, J., & Brunner, M. I. (2024). Hydrological Drought-To-Flood Transitions Across Different Hydroclimates in the United States. *Water Resources Research*, 60(7), e2023WR036504. <https://doi.org/10.1029/2023WR036504>
- Grafton, R. Q., Williams, J., Perry, C. J., Molle, F., Ringler, C., Steduto, P., et al. (2018). The paradox of irrigation efficiency. *Science*, 361(6404), 748–750. <https://doi.org/10.1126/science.aat9314>
- Ho, M., Lall, U., Allaire, M., Devineni, N., Kwon, H. H., Pal, I., et al. (2017). The future role of dams in the United States of America. *Water Resources Research*, 53(2), 982–998. <https://doi.org/10.1002/2016WR019905>
- Kellner, E. (2021). The controversial debate on the role of water reservoirs in reducing water scarcity. *WIREs Water*, 8(3), e1514. <https://doi.org/10.1002/wat2.1514>
- Kim, K. Y., Rajaram, H., & Lakshmi, V. (2025). Observing decreasing snow cover and increasing surface temperature across the Andes with remotely sensed and reanalysis data. *Environmental Research Communications*, 7, 021009. <https://doi.org/10.1088/2515-7620/adb382>
- Lee, C., Kang, S., & Kim, S. (2026). Drought resistance of dams based on propagation analysis: A case study of various multipurpose dams in South Korea. *Journal of Hydrology: Regional Studies*, 63, 103106. <https://doi.org/10.1016/j.ejrh.2026.103106>
- Lema, F., Mendoza, P. A., Vásquez, N. A., Mizukami, N., Zambrano-Bigiarini, M., & Vargas, X. (2025). Technical note: What does the Standardized Streamflow Index actually reflect? Insights and implications for hydrological drought analysis. *Hydrology and Earth System Sciences*, 29(8), 1981–2002. <https://doi.org/10.5194/hess-29-1981-2025>
- Li, Y., Zhao, G., Allen, G. H., & Gao, H. (2023). Diminishing storage returns of reservoir construction. *Nature Communications*, 14(1), 3203. <https://doi.org/10.1038/s41467-023-38843-5>

- 930 López-Moreno, J. I., Vicente-Serrano, S. M., Beguería, S., García-Ruiz, J. M., Portela,
 931 M. M., & Almeida, A. B. (2009). Dam effects on droughts magnitude and du-
 932 ration in a transboundary basin: The Lower River Tagus, Spain and Portugal.
 933 *Water Resources Research*, 45(2), 2008WR007198. [https://doi.org/10.1029/](https://doi.org/10.1029/2008WR007198)
 934 [2008WR007198](https://doi.org/10.1029/2008WR007198)
- 935 Macpherson, E., Clark, C., Salazar, P. W., Baird, N., Akhtar-Khavari, A., & Challies,
 936 E. (2023). Evolving rights to (and of) water in Chile: A case for relationship-based
 937 water law and governance. *The International Journal of Human Rights*, 1–26.
 938 <https://doi.org/10.1080/13642987.2023.2266719>
- 939 McKee, T. B., Doesken, N. J., & Kleist, J. (1993). The relationship of drought
 940 frequency and duration to time scales. *Proceedings of the Ninth Conference on*
 941 *Applied Climatology, American Meteorological Society*, 17(22), 179–184.
- 942 Meza, F. J. (2013). Recent trends and {ENSO} influence on droughts in {N}orthern
 943 {C}hile: {A}n application of the {S}tandardized {P}recipitation {E}vapotranspira-
 944 tion {I}ndex. *Weather and Climate Extremes*, 1, 51–58.
- 945 Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G.,
 946 Balsamo, G., et al. (2021). ERA5-Land: A state-of-the-art global reanalysis
 947 dataset for land applications. *Earth System Science Data*, 13(9), 4349–4383.
 948 <https://doi.org/10.5194/essd-13-4349-2021>
- 949 Oertel, M., Meza, F. J., & Gironás, J. (2020). Observed trends and relationships
 950 between ENSO and standardized hydrometeorological drought indices in cen-
 951 tral Chile. *Hydrological Processes*, 34(2), 159–174. [https://doi.org/10.1002/](https://doi.org/10.1002/hyp.13596)
 952 [hyp.13596](https://doi.org/10.1002/hyp.13596)
- 953 Reade Malagueño, B., & D’Odorico, P. (2024). *¿Libre de la Maleza Estadística?* As-
 954 sessing Neoliberal Promises and Water Markets in Chile. *Earth’s Future*, 12(10),
 955 e2024EF004963. <https://doi.org/10.1029/2024EF004963>
- 956 Rivera, D., Godoy-Faúndez, A., Lillo, M., Alvez, A., Delgado, V., Gonzalo-Martín,
 957 C., et al. (2016). Legal disputes as a proxy for regional conflicts over water

- rights in Chile. *Journal of Hydrology*, 535, 36–45. <https://doi.org/10.1016/j.jhydrol.2016.01.057>
- Rivera-Vidal, R., Arumí, J. L., Melo, O., Delgado, V., Parra, V., Stehr, A., & Daniele, L. (2025). Managed aquifer recharge implementation challenges: Lessons from Chile’s water-scarce regions. *Groundwater for Sustainable Development*, 31, 101502. <https://doi.org/10.1016/j.gsd.2025.101502>
- Scanlon, B. R., Fakhreddine, S., Rateb, A., De Graaf, I., Famiglietti, J., Gleeson, T., et al. (2023). Global water resources and the role of groundwater in a resilient water future. *Nature Reviews Earth & Environment*, 4(2), 87–101. <https://doi.org/10.1038/s43017-022-00378-6>
- Schilstra, M., Wang, W., Van Oel, P. R., Wang, J., & Cheng, H. (2024). The effects of reservoir storage and water use on the upstream–downstream drought propagation. *Journal of Hydrology*, 631, 130668. <https://doi.org/10.1016/j.jhydrol.2024.130668>
- Taucare, M., Viguié, B., Figueroa, R., & Daniele, L. (2024). The alarming state of Central Chile’s groundwater resources: A paradigmatic case of a lasting overexploitation. *Science of The Total Environment*, 906, 167723. <https://doi.org/10.1016/j.scitotenv.2023.167723>
- Vásquez, N. A., Mendoza, P. A., Lagos-Zuñiga, M., Scaff, L., Muñoz-Castro, E., & Vargas, X. (2025). Robust spatial changes in climate classes: Insights from bias-corrected CMIP6 models across Chile. *Environmental Research Letters*, 20(1), 014061. <https://doi.org/10.1088/1748-9326/ad9d5b>
- Vicente-Serrano, Sergio M., Beguería, S., & López-Moreno, J. I. (2010). A multiscalar drought index sensitive to global warming: The standardized precipitation evapotranspiration index. *Journal of Climate*, 23(7), 1696–1718. <https://doi.org/10.1175/2009JCLI2909.1>
- Vicente-Serrano, Sergio M., López-Moreno, J. I., Beguería, S., Lorenzo-Lacruz, J., Azorin-Molina, C., & Morán-Tejeda, E. (2012). Accurate computation of

- a streamflow drought index. *J. Hydrol. Eng.*, 17(2), 318–332. Retrieved from
<http://hdl.handle.net/10261/49224>
- Vicente-Serrano, S. M., McVicar, T. R., Miralles, D. G., Yang, Y., & Tomas-Burguera, M. (2019). Unraveling the influence of atmospheric evaporative demand on drought and its response to climate change. *WIREs Climate Change*.
<https://doi.org/10.1002/wcc.632>
- Wan, W., Zhao, J., Li, H., Mishra, A., Ruby Leung, L., Hejazi, M., et al. (2017). Hydrological Drought in the Anthropocene: Impacts of Local Water Extraction and Reservoir Regulation in the U.S. *Journal of Geophysical Research: Atmospheres*, 122(21). <https://doi.org/10.1002/2017JD026899>
- Wang, Z., Jiang, L., Nielsen, K., & Wang, L. (2024). Reservoir Filling Up Problems in a Changing Climate: Insights From CryoSat-2 Altimetry. *Geophysical Research Letters*, 51(10), e2024GL108934. <https://doi.org/10.1029/2024GL108934>
- Webb, M. J., Winter, J. M., Spera, S. A., Chipman, J. W., & Osterberg, E. C. (2021). Water, agriculture, and climate dynamics in central Chile’s Aconcagua River Basin. *Physical Geography*, 42(5), 395–415. <https://doi.org/10.1080/02723646.2020.1790719>
- Wu, J., Chen, X., Yao, H., Gao, L., Chen, Y., & Liu, M. (2017). Non-linear relationship of hydrological drought responding to meteorological drought and impact of a large reservoir. *Journal of Hydrology*, 551, 495–507. <https://doi.org/10.1016/j.jhydrol.2017.06.029>
- Wu, J., Liu, Z., Yao, H., Chen, X., Chen, X., Zheng, Y., & He, Y. (2018). Impacts of reservoir operations on multi-scale correlations between hydrological drought and meteorological drought. *Journal of Hydrology*, 563, 726–736.
<https://doi.org/10.1016/j.jhydrol.2018.06.053>
- Wu, J., Chen, X., Yu, Z., Yao, H., Li, W., & Zhang, D. (2019). Assessing the impact of human regulations on hydrological drought development and recovery based

- 1013 on a “simulated-observed” comparison of the SWAT model. *Journal of Hydrology*,
 1014 577, 123990. <https://doi.org/10.1016/j.jhydrol.2019.123990>
- 1015 Wu, J., Chen, X., Yu, Z., Yao, H., Li, W., & Zhang, D. (2021). Reservoirs regu-
 1016 late the relationship between hydrological drought recovery water and drought
 1017 characteristics. *Journal of Hydrology*, 597, 126127. [https://doi.org/10.1016/](https://doi.org/10.1016/j.jhydrol.2021.126127)
 1018 [j.jhydrol.2021.126127](https://doi.org/10.1016/j.jhydrol.2021.126127)
- 1019 Zambrano Bigiarini, F. (2026). Reservoir effectiveness and drought vulnerability
 1020 in Chile: Analysis pipeline, reservoir storage records, and result tables. Zenodo.
 1021 <https://doi.org/10.5281/zenodo.21052149>
- 1022 Zambrano, F., Wardlow, B., Tadesse, T., Lillo-Saavedra, M., & Lagos, O. (2017).
 1023 Evaluating satellite-derived long-term historical precipitation datasets for drought
 1024 monitoring in Chile. *Atmospheric Research*, 186, 26–42. [https://doi.org/](https://doi.org/10.1016/j.atmosres.2016.11.006)
 1025 [10.1016/j.atmosres.2016.11.006](https://doi.org/10.1016/j.atmosres.2016.11.006)
- 1026 Zambrano, F., Vrieling, A., Nelson, A., Meroni, M., & Tadesse, T. (2018). Prediction
 1027 of drought-induced reduction of agricultural productivity in Chile from MODIS,
 1028 rainfall estimates, and climate oscillation indices. *Remote Sensing of Environment*,
 1029 219, 15–30. <https://doi.org/10.1016/j.rse.2018.10.006>
- 1030 Zambrano, F., Vrieling, A., Meza, F., Duran-Llacer, I., Fernández, F., Vene-
 1031 gas-González, A., et al. (2025). From Drought to Aridification: Land-Cover
 1032 Fingerprints of a Drying Chile. *Earth’s Future*, 13(12), e2025EF006744. [https://](https://doi.org/10.1029/2025EF006744)
 1033 doi.org/10.1029/2025EF006744